

Project Estimation Techniques.

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- Important project planning Activity
- Different parameters of a project that need to be estimated, include:-

- 1) Project Size
- 2) Effort required to complete the project
- 3) Project Duration
- 4) Cost.

- Important is to accurate estimation of these parameters is important for resource Planning & scheduling.
- Estimation techniques can be classified as:
 - 1) Heuristic Est. Tech.
 - 2) Empirical " "
 - 3) Analytical " "

① Empirical Est. Techniques

③
[Est. of
Cost, time, effort
based on guess/commensense]

based on making an educated guess of the project parameters.

- Technique based on prior experience of development of similar products & projects.
- Guess based on past experience.

1) Expert judgement Technique. → Expert make an educated guess after analyzing the problem.

2) Delphi Cost Estimation

Carried out by team comprising of group of experts.

② Heuristic Technique

The relationship that exist among the different Project parameters can modeled using suitable mathematical expressions.

③ Analytical Estimation Techniques

derive the required results starting with certain basic assumptions regarding a project.

- Certain scientific basis.

COCOMO [Constructive Cost Model] ⑤

Proposed by Dr. Barry Boehm in 1981.

- is heuristic estimation technique, assumes that relationship among different parameters can be modeled using some mathematical expression.
- This approach implies size is primary factor for cost.
- Structures mean complexity.

Three stage process for project estimation

- 1) Line of Code Basis., ranging in size from 2,000 to 100,000 lines of codes.
- 2) Programming languages ranging ~~from assembly to~~
- 3). These projects were based on waterfall Model of software development.

Software development project can be classified into three categories. ⑦

1) Organic , (2) Semi Detached , (3) Embedded

Mode	Project size	Nature of Project	Innovation	Deadline of the project	Development Environment
Organic	Typically 2-50 KLOC	Small size project, experienced developers in the familiar environment. For example, pay roll, inventory projects etc.	Little	Not tight	Familiar & In house
Semi detached	Typically 50-300 KLOC	Medium size project, Medium size team, Average previous experience on similar project. For example: Utility systems like compilers, database systems, editors etc.	Medium	Medium	Medium
Embedded	Typically over 300 KLOC	Large project, Real time systems, Complex interfaces, Very little previous experience. For example: ATMs, Air Traffic Control etc.	Significant	Tight	Complex Hardware/customer Interfaces required

Person Month (PM)

effort of estimation is expressed in units of Person months (PM).

- 1 PM means effort put by person in 1 month.
- As requirement of staff varies ~~for~~ time to time, in the development so there is not constant no. of people is there to work.
- It is calculated from the graph b/w no. of people working & time (in month).

Basic Colomo Model

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Computes software development effort, time and cost as a function of program size.

Program size is expressed in estimated thousands of source lines of codes.

$$\text{EFFORT} = a_1 \times (\text{kLOC})^{a_2} \text{ PM}$$

$$T_{\text{dev}} = b_1 \times (\text{EFFORT})^{b_2} \text{ Months.}$$

Where,

KLOC estimated no. of delivered lines of code for project, estimated size of software product.
(expressed in thousand).

- a_1, a_2, b_1, b_2 are constant for each category of software product.
- T_{dev} is the estimated time to develop the SW in months.
- Effort is the total efforts required to develop the SW product expressed in Person*Months (PM).

Values

Estimation of Development Effort.

$$\text{Organic} = \text{Effort} = 2.4 (\text{kLOC})^{1.05} \text{ PM}$$

$$\text{Semi-detached} = \text{Effort} = 3.0 (\text{kLOC})^{1.12} \text{ PM}$$

$$\text{Embedded} = \text{Effort} = 3.6 (\text{kLOC})^{1.20} \text{ PM}$$

Software Projects	a_1	a_2	b_1	b_2
Organic	2.4	1.05	2.5	0.38
Semidetached	3.0	1.12	2.5	0.35
Embedded	3.6	1.20	2.5	0.32

(12).

Problem:- Consider a software Project using Semi detached mode with 30,000 lines of code. We will obtain estimation for this project as follows:

① EFFORT ESTIMATION

$$E = a_1 (KLOC)^{a_2} PM.$$

$$= 3.0 (30)^{1.12}$$

$$= 135 PM.$$

Where lines of code
30000 = (30 KLOC).

② DURATION ESTIMATION

$$\overline{T_{dev.}} = b_1 \times (\text{EFFORT})^{b_2} \text{ MONTHS}$$

$$2.5 (135)^{0.35}$$

$$D = 14 \text{ months.}$$

③ Person Estimation
Average staff size (ss)

$$N = E/D$$

Estimation/Time.

$$= 135/14 = 10 \text{ Persons approx.}$$

④ Productivity (P) $\frac{\text{KLOC}}{E}$ KLOC/PM)

Ranges (Project size) must be remember to identify mode i.e. Organic, Semidetached, Embedded.

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